

ADVANCED NETWORK SECURITY AND ARCHITECTURES

METI

Report Module 2 [EN]

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1 Introduction

This report describes the configuration and analysis of IPsec VPN using IKEv1 with digital certificates (PKI-based authentication using SCEP). The objective is to replace pre-shared keys with RSA signatures issued by a Certification Authority (CA).

2 Network Setup

The topology consists of routers R1, R2 and RA (Certification Authority). RA also provides NTP synchronization and PKI services.

3 PKI Installation Verification

The following IOS commands confirm successful installation of CA and router certificates:

- `show crypto pki certificates`
- `show crypto ca certificate`
- `show crypto key mypub rsa`

Both R1 and R2 successfully enrolled with the CA and obtained signed X.509 certificates via SCEP. The corresponding RSA public keys were generated locally and included in the certificate exchange process.

4 SCEP Enrollment Analysis

SCEP (Simple Certificate Enrollment Protocol) is used to automate certificate distribution between routers and the CA using HTTP.

The enrollment process consists of:

- CA certificate request
- CA certificate response
- Router certificate request
- Router certificate issuance

The captured HTTP messages show certificate exchange between R1 and the CA. The content-type field indicates certificate payload transfer (.crt).